

BODAYBO, Russian Federation

WMO: 302530

Lat: 57.865N

Lon: 114.239E

Elev: 281

StdP: 98.00

Time Zone: 8.00 (E08)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.1	(c) -42.2	(d) -46.0	(e) 0.0	(f) -43.9	(g) -44.3	(h) 0.0	(i) -42.2	(j) 5.6	(k) -16.5	(l) 4.4	(m) -16.0	(n) 0.1	(o) 200	(p) 0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	30.3	18.8	28.2	18.3	26.2	17.5	20.4	26.7	19.4	25.6	18.5	24.3	2.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.9	22.1	17.3	12.8	21.5	16.2	11.9	21.1	60.1	27.5	56.8	25.3	53.6	24.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	5.0	4.2	-45.2	33.8	1.5	0.6	-46.2	34.2	-47.1	34.6	-47.9	34.9	-49.0	35.3
			-45.3	22.0	1.5	1.9	-46.4	23.4	-47.3	24.5	-48.1	25.6	-49.2	27.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.0	-29.2	-22.4	-13.1	-1.4	6.9	15.2	19.7	16.1	7.0	-3.0	-17.3	-28.2
	DBStd	17.99	8.03	8.49	7.45	4.16	4.26	4.17	3.34	3.32	3.92	5.43	9.22	8.66
	HDD10.0	5818	1216	908	717	342	116	7	1	1	104	404	818	1184
	HDD18.3	8266	1474	1141	976	592	354	108	25	83	340	662	1068	1443
	CDD10.0	691	0	0	0	0	20	164	301	192	14	0	0	0
	CDD18.3	97	0	0	0	0	0	16	67	15	0	0	0	0
	CDH23.3	1155	0	0	0	0	17	272	704	160	1	0	0	0
	CDH26.7	347	0	0	0	0	2	76	234	35	0	0	0	0
Wind	WSAvg	1.4	0.7	1.1	1.6	2.1	2.0	1.8	1.5	1.3	1.5	1.6	1.1	0.7
Precipitation	PrecAvg	447	22	13	9	15	37	65	75	76	58	24	28	25
	PrecMax	597	42	28	25	39	91	130	182	156	130	50	49	46
	PrecMin	334	4	4	2	3	7	2	17	21	25	5	11	11
	PrecStd	76	8	6	5	8	17	26	40	38	23	11	12	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.7	-1.1	3.6	14.2	25.8	31.7	32.9	30.3	21.2	8.6	3.5	-5.3
		MCWB	-7.7	-4.0	-0.3	6.5	14.1	17.8	19.8	18.9	14.3	3.9	1.5	-6.7
	2%	DB	-12.0	-4.6	1.3	9.7	22.0	29.0	31.2	27.1	17.8	6.4	1.1	-8.2
		MCWB	-12.7	-6.0	-1.8	4.3	11.8	17.1	19.3	17.4	11.8	2.6	-0.2	-9.0
	5%	DB	-15.4	-7.3	-0.4	7.1	19.2	26.7	29.3	25.1	15.0	4.7	-1.0	-12.2
		MCWB	-15.9	-8.3	-3.0	2.3	10.3	16.5	18.8	17.7	10.7	2.2	-2.1	-12.8
	10%	DB	-17.8	-10.3	-2.3	5.0	15.7	24.2	27.5	22.9	12.8	3.3	-4.9	-15.4
		MCWB	-18.2	-11.1	-4.6	1.0	8.9	15.5	18.4	16.7	9.8	1.4	-5.9	-15.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.5	-3.8	0.4	6.6	15.2	19.6	21.8	21.2	15.6	5.0	1.8	-6.4
		MCDB	-6.4	-1.6	2.5	12.2	25.4	27.1	29.2	25.2	19.8	6.6	2.9	-5.6
	2%	WB	-12.8	-5.8	-1.5	4.5	12.6	18.4	20.7	19.6	12.8	3.6	0.1	-9.2
		MCDB	-12.0	-4.5	1.4	9.3	20.5	25.8	27.5	23.9	16.1	5.3	1.0	-8.1
	5%	WB	-15.8	-8.4	-3.0	2.8	11.0	17.5	20.1	18.4	11.4	2.5	-2.1	-12.8
		MCDB	-15.3	-7.4	-0.5	6.3	18.6	24.5	26.7	23.4	14.2	4.2	-0.9	-12.1
	10%	WB	-18.2	-11.0	-4.5	1.4	9.1	16.5	19.3	17.3	10.0	1.5	-5.9	-15.7
		MCDB	-17.7	-10.2	-2.5	4.5	14.9	22.8	25.7	22.0	12.6	3.0	-4.9	-15.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	9.3	12.5	10.4	11.9	13.0	12.6	11.2	8.3	6.5	6.9	6.5
		MCDBR	8.8	9.8	10.9	13.4	19.0	18.1	17.7	16.1	13.2	8.4	6.6	8.6
		MCWBR	8.4	8.4	8.4	8.5	9.9	7.8	7.0	7.2	7.6	5.7	5.3	8.0
	5% WB	MCDBR	8.8	9.5	10.5	12.5	17.8	14.9	14.2	13.8	11.7	7.5	6.5	8.5
		MCWBR	8.4	8.1	8.1	8.3	10.0	7.3	6.2	6.7	7.3	5.6	5.5	7.9
Clear-Sky Solar Irradiance	taub		0.256	0.292	0.315	0.357	0.403	0.436	0.447	0.403	0.342	0.300	0.263	0.240
	taud		2.005	2.013	2.042	2.052	2.152	2.217	2.219	2.326	2.436	2.330	2.047	1.964
	Ebn at Noon		563	717	817	843	828	803	783	796	802	728	546	454
	Edh at Noon		58	92	119	140	136	130	127	107	82	67	54	43
All-Sky Solar Radiation	RadAvg		0.68	1.63	3.31	5.05	5.45	5.37	4.99	4.02	2.45	1.34	0.78	0.44
	RadStd		0.04	0.10	0.16	0.24	0.42	0.46	0.54	0.35	0.26	0.16	0.09	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page